

Sameer Gururaj Mathad

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EDUCATION

DePaul University, Chicago, IL — **Masters in Artificial Intelligence [4 GPA]**, September 2023 - June 2025

Global Academy of Technology, Bengaluru, India — **Bachelor in Information Science & Engineering**, August 2020

WORK EXPERIENCE

Assistant System Engineer, Tata Consultancy Services — April 2021 - September 2022

- **Automated SME onboarding and device insurance journeys** by integrating CIPC validation and fraud detection APIs into Siebel workflows, working to achieve **100% automated verification** and eliminate manual vetting, accelerating high-volume customer onboarding.
- **Implemented Oracle MTN staff service journeys** using staff ID-based workflows in collaboration with cross-functional teams, reducing internal search and processing effort by **~25%** and improving operational efficiency for staff services.
- **Built and optimized Oracle SQL-driven backend integrations** between Siebel and external systems, collaborating with teams to improve data reliability and **reduce integration errors by ~20%** across both customer and staff journeys.

PROJECTS

Parser (PERN + LangChain)

- Engineered a production-ready RAG pipeline using LangChain for conversational Q&A over PDFs, capable of handling **multi-page documents** and storing **vector embeddings** for efficient semantic search in PostgreSQL with Pg-Vector.
- Eliminated user wait time for PDF uploads (up to 50MB) by offloading heavy parsing, chunking, and embedding tasks to a **background Worker via Redis**, enabling **near real-time processing for concurrent uploads**.
- Pioneered the full-stack application on the **PERN** stack (Express, React, Node.js, Postgres), securing the service with **JWT** authentication achieving at least **100%** success rate.
- Designed a **multi-service deployment architecture** using Docker Compose, orchestrating API, Worker, Redis, Postgres, and S3, enabling **reliable background processing** and **persistent storage** for user PDFs.

Chat App on Indian History

- Engineered a dual-model (GPT-2) pipeline instruction-tuned with **LoRA** on **14K+** historical Q&A samples, achieving a **77%** training loss reduction (**1.79 -> 0.41**) in the primary model.
- Developed and fine-tuned a secondary model for automated contextual title generation, achieving a **34%** training loss reduction (**1.88 -> 1.23**) with a validation loss of **1.48** on **10K** chat title samples.
- Built a full-stack AI chat app with **React Native frontend**, **Node.js central server**, and **dual FastAPI model servers** for chat and historical Q&A, supporting **modular, scalable real-time inference**.
- Extended **Neural Style Transfer (NST)** to learn composite style representations from abstract art and Indian sculpture images using **VGG19** feature-activations, enabling generative image synthesis and producing **41 unique** UI assets.

English Indic Transformer

- Orchestrated an encoder-decoder Transformer (~**93M**) from scratch in PyTorch, manually implementing **8+ reusable architectural modules** spanning attention modules, feed-forward networks, and decoder causality logic.
- Trained the model on **1.7M English-Hindi sentence pairs**, achieving **~6.3 BLEU after 15 epochs**; trained a **50K SentencePiece Tokenizer** and analyzed translation quality.
- Accelerated auto-regressive decoding by **~40%** through **key-value caching optimization** within the English Indic Transformer project, leading to faster translation times and improved user experience.

Scientific Text Classification

- Led a team-based project to fine-tune DistilBERT on 40K scientific text samples across **10 domains**, achieving a F1-score of **0.86** and **86%** accuracy on a 10K-sample test set, ensuring balanced label distribution across domains to reduce bias.
- Authored end-to-end **CI/CD** with **GitHub Actions**, **CML**, and pre-commit hooks; enabled data/model versioning using **DVC** with remote **AWS S3 integration**, and integrated a **Model Registry** for streamlined **LLMOps** Practices.
- Containerized training/inference pipelines with **Docker** and **Docker Compose**; deployed **FastAPI** backend on **GCP Cloud Run** along with support for **Prometheus-based performance monitoring**.

INDUSTRY SKILLS

Generative AI & LLMs: RAG Pipelines, LLM (Post-Tuning, Pre-Training, Preference Tuning), LoRA Fine-Tuning, LangChain, Transformers, Vision Transformers (ViT), Multimodal LLMs, Prompt Engineering, Model Distillation, Agentic AI (MCP, A2A).

ML/Frameworks: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, NLP, Computer Vision.

MLOps/DevOps: Docker, DVC, Kubernetes, GitHub Actions (CI/CD), MLflow, Prometheus, Hydra.

Full Stack/Databases: PERN/MERN, React, Express, FastAPI, PostgreSQL (PGVector), MongoDB, Redis, ChromaDB.

Cloud/Certifications: AWS, GCP, Azure AI-900 Certified.

Programming Languages: Python, C/C++, TypeScript.